FAIR 3D HERITAGE
Documentation, Publication and Preservation of the Digital 3D Heritage on the Web

Bridging 3D Visualisation and IIIF : The DLF AIM 3D Viewer for Interoperable and FAIR 3D Workflows

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Abstract

This contribution presents the DLF AIM 3D Viewer, an open-source, cross-platform framework for interoperable 3D visualisation and management of heterogeneous digital assets. The work proposes a unified architecture that combines automated multi-format processing, native IIIF interoperability, CMS integration, and extensible interaction mechanisms within a single workflow supporting FAIR principles. The primary objective is to address limitations of existing lightweight viewers in handling heterogeneous datasets, maintaining performance across desktop and mobile devices, and supporting specialised workflows such as inspection, monitoring, annotation, and long-term digital asset management.

The research is motivated by the increasing demand for accessible web-based 3D visualisation that does not depend on proprietary ecosystems, high-end hardware, or tightly coupled infrastructures. Existing solutions such as ATON [1], IIIF-based frameworks [2], Kompakkt [3], and Model Viewer [4] each address different aspects of web-based 3D interaction. ATON focuses on scalable infrastructures for complex environments, IIIF initiatives promote interoperability and standardised access to digital resources, Kompakkt introduces collaborative annotation and storytelling capabilities, while Model Viewer provides a lightweight component for embedding 3D content within web applications.

However, existing approaches reveal several limitations. End-to-end processing pipelines capable of transforming heterogeneous 3D formats into a consistent, optimised representation remain limited. Furthermore, many systems prioritise either simplicity or advanced functionality without providing an extensible architecture suitable for integration with custom repositories and automated workflows. Support for deployment across content management systems, standalone environments, and embedded applications is also often constrained, reducing flexibility for institutions managing diverse digital collections.

The proposed system is implemented primarily in JavaScript using the three.js library [5] for real-time rendering. It is distributed as an open-source Drupal module [9] and is publicly available through GitHub [10], while remaining independent of any specific platform. The viewer can therefore be deployed within CMS environments, as a standalone web application, sandbox mode, in offline installations, or embedded into external web applications through a dedicated integration mode. This deployment flexibility enables adoption across diverse institutional infrastructures without requiring modifications to existing repository architectures.

Fig. 1 Overview of the AIM 3D Viewer workflow and processing pipeline.

The client application is complemented by a server-side processing environment implemented in PHP and Bash. The system supports numerous 3D formats, including OBJ, DAE, FBX, PLY, IFC, STL, XYZ, JSON, 3DS, and glTF/GLB [6]. Uploaded models may be automatically decompressed, converted into an optimised GLB representation using Blender-based processing scripts [7], and enriched with generated thumbnail previews. Beyond format conversion, the processing pipeline normalises heterogeneous source data into a unified representation, simplifying publication, reuse, and long-term management while ensuring consistent rendering behaviour across supported formats.

Native compatibility with IIIF Manifests [2] enables structured loading of complete 3D scenes using standardised metadata and external resource references. Rather than serving solely as a transport mechanism, IIIF manifests preserve scene configuration, metadata associations, and links to distributed resources, enabling interoperable and reproducible workflows across IIIF-compliant repositories and applications. This work proposes the AIM3IDF Manifest (AIM 3D Interoperability Format), an extension of the

IIIF Manifest specification that combines standardised IIIF metadata with DLF AIM 3D Viewer configuration, enabling portable and reproducible 3D scene descriptions while preserving interoperability with IIIF-compliant repositories. This approach supports FAIR principles by improving the accessibility, interoperability, and reusability of digital 3D assets while facilitating integration with existing cultural heritage infrastructures.

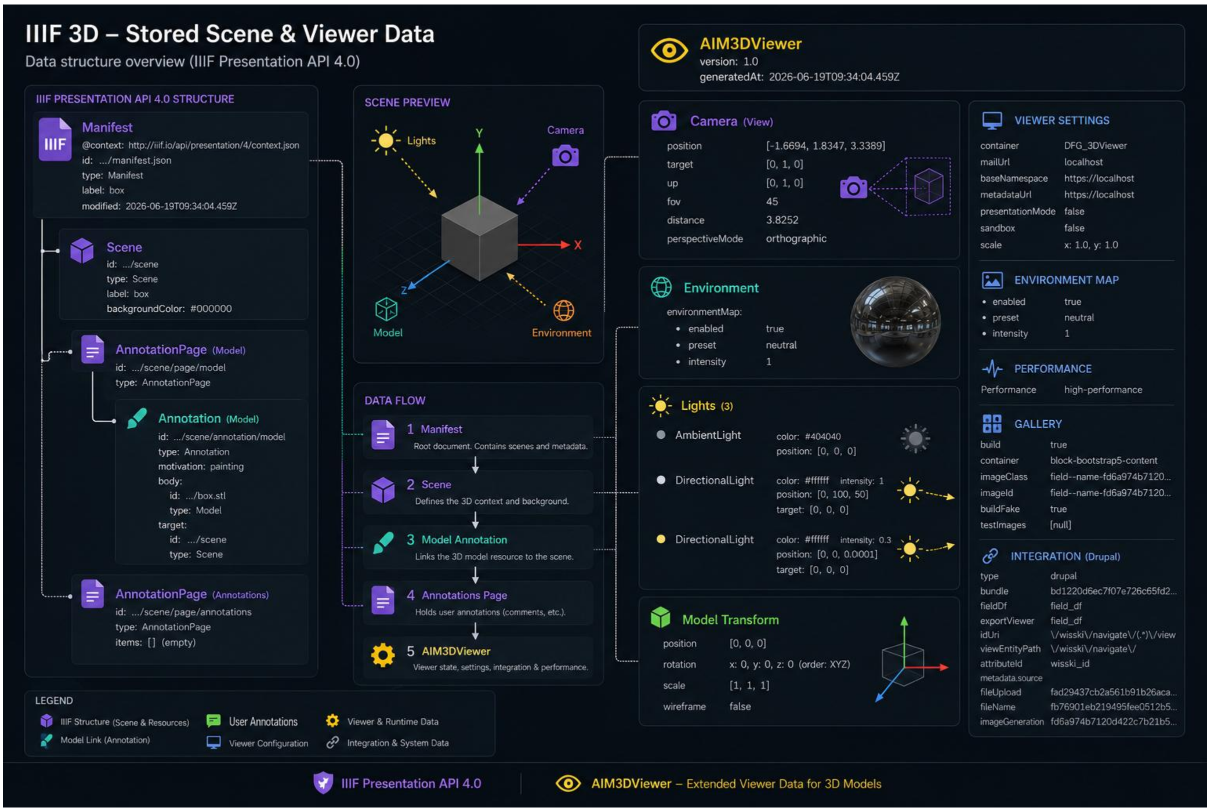

Fig. 2 Overview of the AIM 3D Viewer workflow and processing pipeline.

The viewer provides interactive tools for object transformation, hierarchical selection, metadata visualisation, measurement, clipping planes, face picking, material management, and persistent scene configuration. These capabilities support practical workflows including heritage object inspection, engineering model validation, collaborative review, and annotation. For example, cultural heritage institutions may automatically transform archival 3D datasets into standardised GLB assets published through IIIF manifests while preserving associated metadata and scene configuration for long-term access and reuse.

The proposed architecture was evaluated using heterogeneous datasets representing multiple supported file formats and varying model complexity. Testing focused on rendering performance, workflow automation, interoperability, and deployment across desktop and mobile environments. Preliminary evaluation demonstrated reliable visualisation of heterogeneous datasets across different platforms while reducing manual preparation effort through automated preprocessing and unified asset conversion. Current scalability is primarily constrained by browser and GPU memory limitations for extremely large models, representing an area for future work involving progressive streaming and level-of-detail techniques.

The DLF AIM 3D Viewer demonstrates how automated processing pipelines, open standards, and IIIF interoperability can be combined within a unified architecture supporting sustainable and FAIR-compliant management of heterogeneous 3D assets. The proposed approach contributes practical solutions for interoperable web-based visualisation and digital preservation workflows applicable to cultural heritage, engineering, digital asset management, education, and digital twins.

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